National University of Computer & Emerging Sciences (FAST-NUCES)

Department of Computer Science — Chiniot-Faisalabad Campus

CL3001 Machine Learning - Lab | Semester Project (Spring 2026)

Instructor: Esha Iftikhar

DELIVERABLE 1: RESEARCH PAPER APPROVAL FORM

Field	Details
Course Code & Title	CL3001: Machine Learning - Lab (Spring 2026)
Student Name(s)	Aqeel Ahmad, Naveed Mehdi, Rahim Ijaz
Roll Number(s)	24F-0033, 24F-0037, 24F-0045
Submission Date	November 2026
Project GitHub URL	https://github.com/aqeelahmad0105-dev/poker-ai-game

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1. Selected Research Paper Details

- **Paper Title:** *Comparison of the Effectiveness of AI Models in Poker Games*
- **Authors:** Mateusz Tustanowski and Adrian P. Wozniak
- **Institutional Affiliation:** Military University of Technology, Warsaw, Poland
- **Publication Venue:** *Communications of International Proceedings*, Vol. 2025 (20), Article ID 4534725
- **Conference Presentation:** 45th IBIMA International Conference, 25–26 June 2025, Cordoba, Spain
- **Publication Date:** June 2025 (Meets 2025/2026 mandatory requirement)
- **Official Paper Link:** https://ibimapublishing.com/p-articles/45AI/2025/4534725/
- **Digital Object Identifier (DOI):** https://doi.org/10.5171/2025.4534725

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2. Dataset Information

- **Dataset Name:** No-Limit Texas Hold'em Multi-Stage Decision & Gameplay Dataset
- **Source / Repository:** Public No-Limit Texas Hold'em game history records (derived from open poker hand log archives)
- **Dataset Link:** Included in project repository: `data/poker_gameplay_paper_data.csv`

- **Dataset Size:** 120,000 game state instances across 14 numerical features and 3 target decision classes (`Fold = 0` , `Check/Call = 1` , `Bet/Raise = 2`).

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3. Justification & Alignment with Course Objectives

1. **Application-Based & Real-World Domain:** Poker represents an iconic stochastic imperfect-information benchmark in artificial intelligence and game theory, modeling real-world decision-making under uncertainty, hidden state variables, and economic risk management.
2. **Core Course Topics Applied:** The paper directly utilizes topics taught in the Machine Learning curriculum: * **Supervised Learning & Ensembles:** Random Forest classifiers, decision tree depth tuning, bagging, feature split criteria, and multi-seed variance analysis. * **Neural Networks & Reinforcement Learning:** Multi-Layer Perceptrons, Deep Q-Networks (DQN), Adam optimizer, ϵ-greedy exploration policies, experience replay buffers, and target networks. * **Model Evaluation & Diagnostics:** Accuracy, F1-scores, confusion matrices, economic return (chips won/lost), and head-to-head tournament simulation.
3. **Reproducibility with Available Resources:** The computational scope is fully reproducible using standard Python data science libraries (`scikit-learn` , `numpy` , `pandas` , `joblib` , `streamlit`) within standard hardware constraints without requiring multi-GPU server clusters.
4. **Feasibility of Meaningful Bonus Improvement:** The paper concludes with an open research recommendation: combining supervised Random Forest baselines with adaptive RL, which we have directly implemented into a full-fledged interactive multiplayer web application.

Instructor Approval Status: _____
Instructor Signature: _____
Date: _____